Lightweight Meets Complete: A Hierarchical Progressive Fusion Network Based on Kolmogorov–Arnold Networks for Hyperspectral Image Classification

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***Abstract*— Existing mainstream hyperspectral image (HSI) classification frameworks, such as multilayer perceptron (MLP) networks, combine learnable linear transformations with non-learnable nonlinear activation functions. While these feature extraction algorithms achieve high classification accuracy, they require significant computational and memory resources, and many models lack adequate informative representations. Recently, fast Kolmogorov–Arnold networks (FKANs) have emerged as a lightweight alternative to MLPs, offering competitive performance with fewer parameters and lower computational costs. This article presents a hierarchical progressive fusion network (HPFN) based on FKAN to address these challenges. The proposed network explores HSI information more comprehensively through branches at three scale levels (pixel, patch, and global) while remaining lightweight. The orthogonality of information at different levels is preserved to some extent for diverse practical applications. We develop two FKAN-based lightweight convolutional networks to extract pixel- and patch-level features. In the global-level branch, a boundary-enhanced U-Net with an edge-constrained module based on Sobel operators mitigates oversmoothing. The cross-scale guided fusion mechanism allows for dynamic awareness and accurate pairing of pixel-level features with local and global features, reducing information redundancy. Experimental results show that the proposed network achieves competitive classification accuracy on four public datasets at low computational cost, validating its effectiveness.**

***Index Terms*— Complete information representation, cross-scale, fast Kolmogorov–Arnold network (FKAN), hyperspectral image (HSI), lightweight.**

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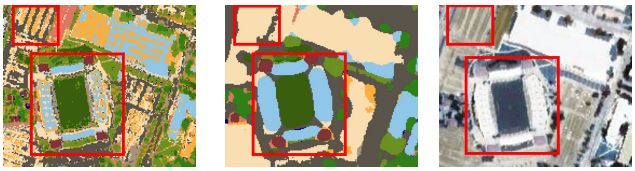
I. Introduction

HYPERSPECTRAL remote sensing imaging is an important Earth observation technique that can provide insight into the complex optical properties of different ground-based objects. This unique description method combined with the spatial features of objects enables hyperspectral image (HSI) to achieve high-precision classification [1], [2]. As a result, HSI classification techniques have become a research hotspot in remote sensing and are widely used in many fields, such as ecological monitoring [3], urban planning [4], smart agriculture [5], and mineral exploration [6], [7].

Since the HSI classification task belongs to pixelwise classification, early research focused on exploiting spectral information. Researchers have developed a variety of related algorithms, such as support vector machine (SVM) [8] with pixel-by-pixel spectral vectors as input, multilayer perceptron (MLP) [9], and stacked autoencoder (SAE) [10]. However, these methods largely ignore the exploration of spatial information, leading to limitations in classification performance. With an emphasis on spatial features such as the distribution and aggregation of ground-based objects, a growing number of studies have started to incorporate spatial information into classification algorithms. For example, inspired by the field of RGB image processing, several studies have used principal component analysis (PCA) [11] to reduce HSI to three channels and extract spatial information via convolutional neural networks (CNNs) [12] for high-precision classification, validating the important role of spatial information in improving classification performance. However, overreliance on spatial information may lead to oversmoothing in classification maps, especially in practical applications, where all pixels of the captured region need to be classified. The classification results of edge pixels between background and object pixels are further affected by spatial information. As shown in Fig. 1, overstrengthening the spatial information causes the target pixel region in the classification result to cover the surrounding background pixels. Thus, how to balance the weight of spatial and spectral information becomes an important challenge in HSI classification tasks.

The convolution kernel of 3-D CNN performs sliding convolution in both spatial and spectral dimensions on the feature

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(a) (b) (c)

Fig. 1. Visual effects of the oversmoothing problem in HSI classification maps. (a) Emphasis on spectral information. (b) Emphasis on spatial information. (c) Fake color image.

maps, which enables the model to adaptively explore the weights of the spatial and spectral information of the HSIs. To further improve the model performance, researchers have introduced various optimization strategies into 3-D CNN [13], [14], such as spatial and spectral attention mechanism [15], [16], skip connections [17], [18], and deformable convolutions [19], [20]. In addition, some models explore the underlying relationship between the target object and the background by fusing local and global information [21], [22], taking into account the distributed nature of objects in remote sensing images. With the development of deep learning, models such as the modified U-Net [23], the graph convolutional network (GCN) [24], [25], and the vision transformer (ViT) [26], [27], [28] have gradually become important methods for HSI classification due to their superior performance in spectral–spatial feature extraction.

However, incorporating global information presents new challenges in balancing spatial and spectral information. Different practical scenes may require varying emphases on these aspects. For instance, in farmland planning, spatial information is crucial for identifying regional distributions, as is evident from ground-based object aggregation. Conversely, in complex scenarios such as urban planning, the clear boundaries of ground objects become more important than the presence of some misclassified points within the same labeled region. Relying heavily on spatial information can blur target edges or obscure surrounding objects, necessitating a stronger emphasis on spectral information. Nonetheless, most existing models use cross-entropy loss functions that are primarily aimed at the classification accuracy of labeled samples, which is difficult to cope with the different requirements of background pixels in practical applications. At the same time, the introduction of additional global information as a form of spatial information will further exacerbate the risk of oversmoothing [29]. While datasets with complete pixel labels can partially alleviate this issue, it is still necessary to further enhance the orthogonality between spectral information and spatial information (especially global and local) in the current conditions. This enhancement of orthogonality facilitates the model to assign spatial–spectral weights on demand to cope with different application scenarios. Therefore, there needs to be a balancing mechanism between autonomous learning and human regulation in HSI classification tasks. Specifically, the model should not only adaptively explore the contribution of spatial–spectral information but also allow flexible tuning of the spatial–spectral weights according to the requirements of different scenes without significantly degrading the performance. This requires the model to balance the recognition ability of the target pixels with the discrimination ability of the background pixels.

Better information collaboration is a promising approach to alleviate these limitations. Spatial information and spectral information, together with local and global feature interactions, are keys to the performance of many HSI classification models. Combining these information types through specific fusion mechanisms can overcome the drawbacks of relying on a single source. Yang et al. [30] proposed a feature reuse strategy to highlight representative features while leveraging more comprehensive information, allowing spatial features to supervise the spectral signal. Similarly, Guo et al. [31] introduced an integrated U-Net and bidirectional LSTM network to capture correlations between global and spectral features. The strong orthogonality between global- and pixel-level information contributes to the high performance of the model in the full-pixel classification task. However, independence limits the interaction of different features, potentially limiting further accuracy improvements.

Moreover, there is a certain restrictive relation between the complete information representation [30] and the lightweight of the model. The information of HSI data can be represented in various forms, such as single-pixel spectral vectors, local–global information, and spatial–spectral- and frequency-domain information. With the continuous development of multimodal learning paradigms [32], [33], [34], light detection and ranging (LiDAR) data [35], [36] and synthetic aperture radar (SAR) data [37] have also been introduced into HSI classification, providing more algorithms and frameworks for multi-input fusion models. In order to fully extract the features of various information, researchers not only need to add more modules to cope with different kinds of information but also need to add some feature enhancement or generalization strategies to each module to avoid potential overfitting and Hughes phenomenon [38]. Many existing deep learning models are based on MLP, that is, the base unit consists of learnable linear transformations and nonlearnable nonlinear activations. While this feature extraction method achieves good performance, it also needs a deeper model and many inefficient neuron nodes [39]. Multiple information processing modules also enhance the model complexity, which further increases the computational cost and memory cost requirements.

In addition to the complexity of the model, MLP-style deep learning models are still affected by some stubborn diseases to varying degrees, such as catastrophic forgetting [40] and poor interpretability [41]. Recently, the Kolmogorov–Arnold network (KAN) [42] has started to appear in various areas of deep learning as an alternative to MLP. KANs use multiple nonlinear spline functions to approximate the mathematical analysis of the model, instead of linear transformations and activation functions in MLPs. In function simulation and classification tasks, KAN achieves competitive classification results while significantly reducing model depth and width. To further improve the computational efficiency of KAN, several works optimize nonlinearities and their approximation procedures, such as efficient KAN (EKAN) [43] and fast KAN (FKAN) [44]. On this basis, Jamali et al. [45] introduced EKANs into the HSI classification task, studied

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the exploration ability of KAN for spatial–spectral information and the classification accuracy under different structures, and verified that EKANs with a more lightweight structure can achieve competitive classification performance compared with advanced models. However, in the field of HSI classification, KANs still lack a more complete feature optimization structure and a more complete information representation than CNNs. This may be one of the main reasons that limit the accuracy of KAN classification.

In this article, based on the above constraints and challenges, we propose a hierarchical progressive fusion network (HPFN). This framework uses FKAN-based convolution to extract patch-level local features and pixel-level spectral features, respectively. FKAN ensures the lightweight nature of HPFN by replacing redundant linear convolutions with nonlinear convolutions with fewer network layers and nodes. This structure enables the model to better balance efficiency, completeness, and accuracy. In addition, we develop a 3-D convolutional edge constraint U-net (ECU-Net) to capture global features. Finally, the designed cross-scale guided fusion mechanism (CSGFM) achieves the precise interaction of various information and preserves to some extent the manual assignment of weights for each. The FKAN algorithm is used to design a convolutional network for feature extraction with low computational and memory costs, and a small number of MLP-style 3-D convolutional operations are used in ECU-Net to achieve global information extraction, thus balancing the complete information representation and model lightweighting. Our main contributions are summarized as follows.

1) We propose an HPFN for cross-scale interaction in HSI classification. Our model consists of a pixel-level spectral information extraction network (Pixel-Net), a patch-level local information extraction network (Patch-Net), and an ECU-Net. The Pixel-Net and Patch-Net branches enable a more lightweight and efficient exploration of spatial–spectral information through the convolutional approach of FKAN. The global information is extracted by ECU-Net with 3-D CNN.
2) A CSGFM is designed as an important innovation of this work. This module enables accurate feature registration and deep fusion from the pixel level and the patch level to the global level. The contribution of different types of information to the model is explored adaptively through learnable parameters, and artificial free allocation margins are reserved for each branch in the final fusion to cope with different task requirements.
3) To alleviate the oversmoothing problem caused by the introduction of multiple spatial information, the Sobel operator is used to design a multidirectional edge constraint module (ECM). Edge enhancement and background suppression are performed on the reduced HSI images to obtain boundary information of different objects, which is used as a new supervised signal to achieve quadratic constraints for the ECU-Net network.

II. Related Works

KANs not only maintain competitive classification accuracy while being more lightweight but also enhance the interpretability of deep learning to some extent. Its unique approximation approach also alleviates the catastrophic forgetting that occurs in MLP-style models.

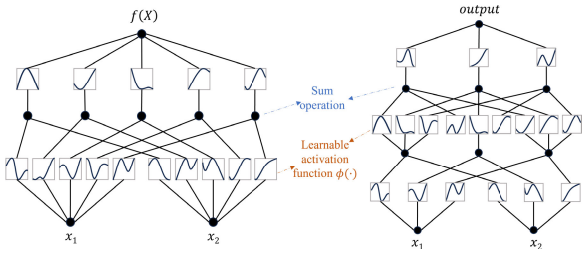


Fig. 2. Structure of two-layer KAN and multilayer KAN.

Equation (1) is the mathematical form of the Kolmogorov–Arnold representation theorem. This theorem shows that any multivariate continuous function $f(x)$ can be represented as a two-level nested superposition of univariate continuous functions

$$f(X) = f(x_1, \ldots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}(x_p) \right) \quad (1)$$

where $X = [x_1, \ldots, x_n]$ represents the input features of n nodes, and $\Phi_q(\cdot)$ and $\phi_{q,p}(\cdot)$ denote the qth outer univariate continuous function and the pth inner univariate function of this function, respectively. As shown in Fig. 2, this mathematical theorem can be constructed as a two-layer neural network. A nonlinear activation function based on a B-spline function is used as a univariate continuous function for this network. N corresponds to the number of neurons fed into the network. Nodes represent the addition operations of each function, and edges represent the flow directions of the function information. In order to smoothly and accurately fit more complex functions, deeper KAN is often required. The sum of multiple inner univariate continuous functions is fed into a different outer function, and the output of the outer function is treated as a new inner function, which enables the construction of an arbitrary layer KAN.

Approximation procedures for spline functions are the core algorithms of KAN. The expression for the B-spline function is given as follows:

$$\text{spline}(x) = \sum_i c_i B_i(x) \quad (2)$$

where $B_i(\cdot)$ denotes the ith spline basis function and c_i denotes the weight of $B_i(\cdot)$. The B-spline function approximates the transformation function at different layers by a linear combination of multiple basis functions. In KAN, the learnable parameters of the model are mainly focused on the coefficients in the linear combination process of these basis functions. Namely, the core of KAN learning is how to optimize the coefficients of each basis function to better fit the target transformation and, thus, improve the expressive power of the model.

In addition, to alleviate the learning pressure on the network, other nonlinear functions such as SiLU functions are integrated into the B-spline function to form the residual activation

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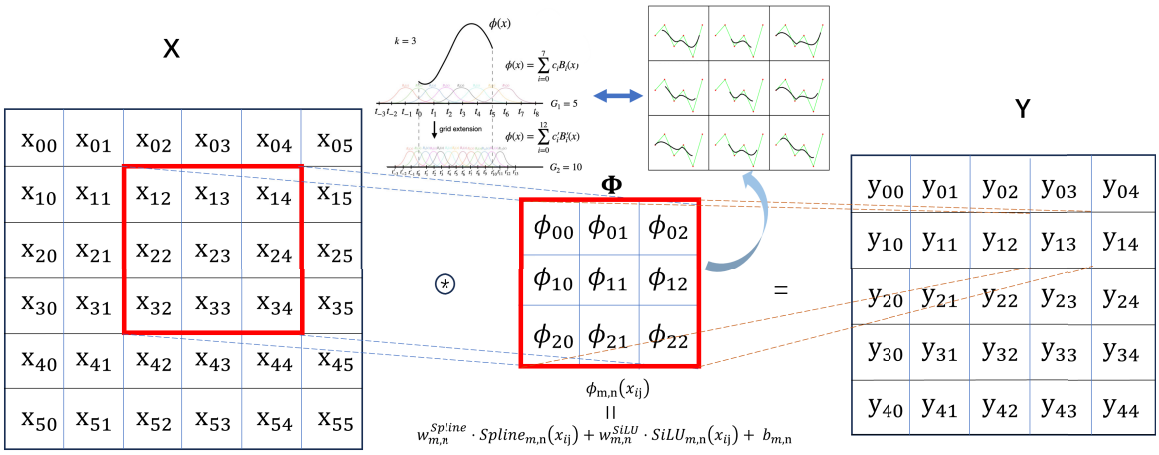


Fig. 3. KAN-based convolution operation process. X and Y denote the input and output features, respectively; ϕ denotes the convolution operator based on KAN. $w_{m,n}^{\text{Spline}}$ and $w_{m,n}^{\text{SiLU}}$ denote the learnable weights of the spline function and SiLU, respectively. (m, n) denotes the index of the KAN-based convolution kernel.

function of KAN. The mathematical statement is given as follows:

$$\phi(x) = w(\text{SiLU}(x) + \text{spline}(x)) \tag{3}$$

where w is used to control the range of $\text{SiLU}(x)$ and $\text{spline}(x)$.

KAN generalizes the Kolmogorov–Arnold representation theorem to arbitrary width and depth, that is, to support an arbitrary number of nodes in an intermediate network layer, and relaxes the original restriction of using only two layers. The propagation process from layer l to layer l+1 in KAN is described as follows:

$$\mathbf{X}_{l+1} = \mathbf{\Phi}_l \mathbf{X}_l = \begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,2,1}(\cdot) & \cdots & \phi_{l,n_{l+1},1}(\cdot) \\ \phi_{l,1,2}(\cdot) & \phi_{l,2,2}(\cdot) & \cdots & \phi_{l,n_{l+1},2}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,1,n_l}(\cdot) & \phi_{l,2,n_l}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix} \mathbf{X}_l \tag{4}$$

where $\mathbf{X}_l = [x_{l,1}, x_{l,2}, \ldots, x_{l,n_{l+1}}]$ denotes the feature vector of the lth layer, n_l denotes the number of nodes in lth layer, and $[\phi_{l,1,i}(\cdot), \phi_{l,2,i}(\cdot), \ldots, \phi_{l,n_l,i}(\cdot)]$ denotes the n_l residual activation function corresponding to the input vector of $x_{l,i}$ in the lth layer ($i \in [1, n_{l+1}]$).

In implementation, B-spline functions often need to be computed via deBoor-Cox iterations, which makes KAN with stronger nonlinear representations a bottleneck in terms of operational efficiency. Since all activation functions used are linear combinations of a fixed set of third-order B-spline basis functions, the efficient implementation of FKAN abstracts the computation into linear combinations of activations after different basis functions [43]. This reduces the process to a basic matrix multiplication, leading to a more lightweight and efficient implementation. Furthermore, Li [44] proposed FKAN, which approximates the third-order B-spline basis function by a radial basis function (RBF) with a Gaussian kernel. By incorporating residual connections, FKAN effectively reduces the computational cost without compromising the performance.

Due to its potential as an alternative to MLP, several studies have started to explore the performance of KAN-based algorithms in more complex deep learning models, such as KAN-based GCNs [46] and CNNs [47]. As shown in Fig. 3, the convolution operation in KAN is similar to that in CNNs, where both perform sliding summation operations over the input tensor. The key difference is that the convolution kernels of CNNs apply linear transformations to each tensor in the window, whereas the convolution operation of KANs incorporates both nonlinear and linear transformations.

To explore the performance of KAN in HSI data processing, Jamali et al. [45] proposed a simple multidimensional convolutional hybrid network based on KAN (HKAN) and explored the classification performance and computational cost of the model in different dimensions. However, HKAN has not fully addressed the efficiency bottleneck of KAN and complete information representation. The potential of KAN-based convolutional methods in extracting spatial–spectral features has not been fully realized. To address this issue, we develop deep learning models based on FKAN at the pixel level, the local level, and the global level. This model balances a lightweight architecture with more comprehensive feature extraction, enabling accurate feature alignment and fusion across scales.

III. Method

In this section, the current mathematical principles of FKAN-convolution are first introduced, based on which the multivision fusion network framework and key modules are described.

A. FKAN-Convolution

FKAN approximates the third-order B-spline basis by a Gaussian RBF [44]. The RBFs between the neural nodes and each grid node are constructed by setting the number of grid points to be evenly divided. Similar to KAN, FKAN also incorporates the $\text{SiLU}(\cdot)$ into its propagation. The propagation formula is given as follows:

$$\begin{aligned} \mathbf{x}_{l+1} &= \mathbf{\Phi}_l^{\text{FKAN}}(\mathbf{x}_l) \\ &= \mathbf{W}_{ls} \cdot (\phi_{\text{RBF}}(\|NL(\mathbf{x}_l) - \mathbf{c}_i\|)) \\ &\quad + \mathbf{b}_{ls} + \mathbf{W}_{lb} \cdot \text{SiLU}(\mathbf{x}_l) + \mathbf{b}_{lb} \end{aligned} \tag{5}$$

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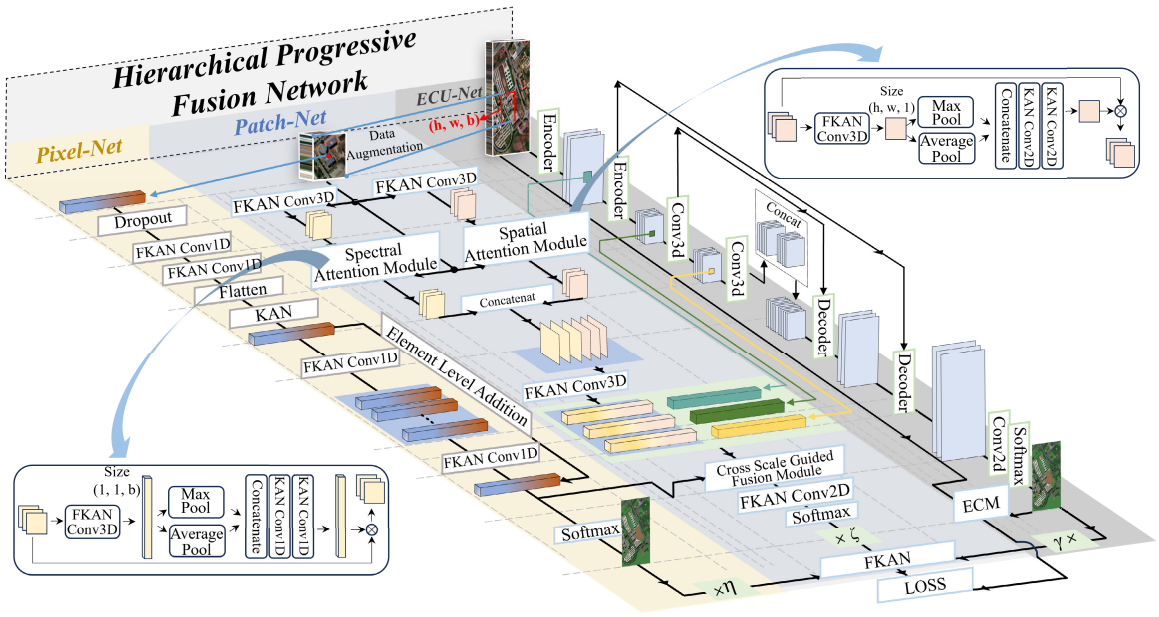


Fig. 4. Structure of the HPFN. The framework contains feature extraction branches at three scales, from left to right: Pixel-Net, Patch-Net, and ECU-Net. η, ζ, and γ denote the weights of the three branches, respectively. h, w, and b denote the height, width, and bands of the patch sample, respectively.

where $\mathbf{W}_{ls}$ and $\mathbf{b}_s$ are the learnable weight and bias of the RBF of the lth layer, respectively; $\mathbf{W}_{lb}$ and $\mathbf{b}_{lb}$ are the learnable weight and bias, respectively. l denotes the index of the current layer. NL$(\cdot)$ denotes the normalization operation. As shown in (5), FKAN approximates a third-order B-spline function by learning a $\mathbf{W}_{ls}$, which is a weighted sum of distances from different grids in the RBF. Similarly, the range of activation functions can be controlled adaptively with trainable $\mathbf{W}_{lb}$ for better adaptation with RBF. Corresponding to the convolution process of KAN, the mathematical formulas for 3-D convolutions based on FKAN are given as follows:

$$\left(\mathrm{X} * W^{\mathrm{FKAN}}\right)_{i,j,k} = \sum_{t=1}^{T}\sum_{n=1}^{N}\sum_{m=1}^{M} \phi_{tnm}^{\mathrm{FKAN}}\left(x_{i+m,j+n,k+t}\right) \quad (6)$$

where $\phi_m^{\mathrm{FKAN}}(\cdot)$ represents the mth function in the function matrix $\mathbf{\Phi}^{\mathrm{FKAN}}$; M, N, and T are used to represent the dimensions of the convolution kernel. W^{FKAN} denotes the convolution kernel of KAN.

B. Overall Framework: HPFN

The proposed model structure is shown in Fig. 4, which includes Pixel-Net, Patch-Net, and ECU-Net. The Pixel-Net and Patch-Net use FKAN-based convolution operations to extract spectral and spatial features, respectively. The ECU-Net uses a U-net with convolution operations of MLP-style and a multidirectional edge constraint algorithm based on Soble operators to alleviate the oversmoothing problem. HPFN adopts a more complete representation of the information and achieves accurate cooperation among various information through the proposed CSGFM. The information interaction of the three branches is divided into intermediate feature fusion and final fusion. In the intermediate fusion process, ECU-Net and Patch-Net achieve feature information interaction through accurate pairing, which alleviates the potential redundancy and conflict risk in the fusion between different network features. On this basis, Pixel-Net realizes the information aggregation of the three branches through an indirect attention fusion mechanism (i.e., CSGFM). In the final fusion stage, the fully connected layers of the three branches are fed into a single-layer FKAN according to their weights to achieve total fusion and final classification. The algorithm for each component of the model is described in detail in the following.

C. Global Feature: ECU-Net

Global features often require that the network can support a full image input in a single batch. Common approaches include GCNs and U-Net architectures. The former requires all pixels or superpixels to be constructed in a sample of a single graph structure, which requires high hardware memory cost. The latter involves convolving the entire HSI, and its computational complexity is directly related to the number of convolution kernels and layers. Currently, some U-Net architectures for HSI classification use deep structures and a large number of channels to fully extract the intrinsic features. This approach has a large negative impact on the efficiency of the model while improving the accuracy. Although KAN-based convolutional algorithms can replace the MLP-type convolution operations in traditional U-net with a smaller network structure, this replacement method does not significantly improve the model efficiency due to the need for global convolution and the high computational complexity of KAN.

Considering the orthogonality and cooperation between global and other information (different branches), we use MLP-style 3-D convolutions to build a lightweight ECU-Net with fewer channels and layers. The proposed network is designed to learn global spatial information with fewer parameters, while spectral information and local information are complemented by other branches. In this way, the proposed

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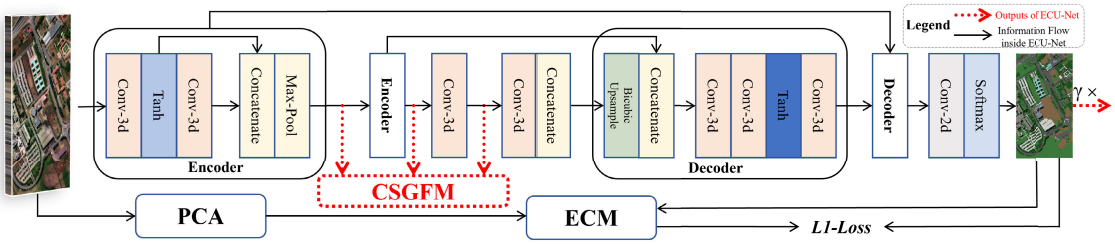


Fig. 5. Detailed structure of ECU-Net. The solid line represents the information flow for ECU-Net, and the dashed line represents the information flow for CSGFM in this network.

model can alleviate the information loss of shallow network structures and also complete the exploration of global information with high computational efficiency.

The structure of the ECU-Net is shown in Fig. 5. This branch of the network contains two encoders and two decoders. The encoder outputs two feature maps, which are used at the subsequent encoder and corresponding decoder. When the input feature map is encoded twice, it is fed into an intermediate layer for further feature extraction. Similar to the encoder structure, the intermediate layer also contains two convolutional layers with the same number of channels as the encoder, and the results are concatenated by the channels. In addition, the decoder contains a bicubic upsampling operation and three single-channel 3-D convolutional layers. This structure ensures that the model is lightweight and mitigates information loss during decoding.

Moreover, global feature extraction is prone to model overdependence on spatial information, leading to oversmoothing phenomena. As the number of learnable parameters decreases, the discriminative ability of the network for background pixels is further limited, which may exacerbate the smoothness of the classification map. To alleviate this large-scale boundary coverage problem, we design an ECM based on multidirectional Sobel operators and incorporate it into the propagation process and loss function, which imposes a quadratic constraint on the output image of the global branch. In contrast to other filtering algorithms, the Sobel operator extracts the intensity and direction of edge changes while accounting for efficiency. Through the combination of multiple custom convolution kernels and specific activation functions, ECM enables fine edge detection and enhancement of HSIs. Thus, the loss function of HPFN includes the cross-entropy loss function and the L1-loss function, which is mathematically formulated as follows:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log\left(\hat{y}_i\right) + k_{L1} \cdot \mathcal{L}_{L1}(I_R, I_P) \tag{7}$$

where $\mathcal{L}$ and $\mathcal{L}_{L1}(\cdot)$ are the total loss function and the L1-loss function, respectively; y_i and $\hat{y}_i$ are the ground truth and prediction results for the ith sample, respectively; C denotes the number of classes; and k_{L1} is used to control the range of the loss function. Moreover, k_{L1} is automatically attenuated to take into account the risk that the L1-loss may affect the classification accuracy. That is, k_{L1} decays by a factor of β whenever the cross-entropy loss increases continuously (n_{L1} iterations).

The detailed algorithm of ECM is shown in Fig. 6. The module first performs edge detection in multiple directions on the input image through several predefined convolution kernels, including the standard Sobel kernel (vertical and horizontal) and the custom diagonal convolution kernel. The proposed algorithm is able to capture more edge information in more directions, thus improving the comprehensiveness and accuracy of model edge detection. In addition, we smooth the extracted edge information with a Gaussian filter to reduce noise interference and further construct a bandpass filter with a series of activation functions for translation and nonlinear transformation to enhance edge saliency and contrast. Finally, the stability and consistency of the output features are guaranteed by the normalization. Subsequent experiments show that the proposed module improves the edge detail representation and effectively mitigates the oversmoothing of the global-level branch.

D. Local Feature: Patch-Net

In the field of remote sensing HSI classification, using patch samples as input to the network is a common approach for most deep learning models. Patch samples can not only feed spatial context information into the network but also preserve the spectral information of the target. Many deep learning models for patch processing use 3-D CNN. This operation of 3-D spatial sliding convolution further reduces the memory cost and explores the assignment of spatial and spectral information weights in a learning manner. However, limited by the high-dimensional nature of HSI data, complex 3-D CNNs are susceptible to the curse of dimensionality and the Hughes effect. It is difficult to efficiently exploit the underlying nonlinear information in complex data. Convolutional operations based on FKAN allow the network to focus more on fitting nonlinear processes. The RBF is used to approximate the spline function, which provides a basis for model lightweightness and avoids the overfitting problem caused by deep models. Therefore, a convolutional network branch based on FKAN is added to the proposed framework to process the patch-level sample information.

As shown in Fig. 4, in this branch, we introduce the spatial attention module and the spectral attention module based on the convolution operation of FKAN, respectively. The designed branch can focus on the effective information in the complex data. The output feature maps are stacked and fed into a convolutional layer to generate feature vectors for

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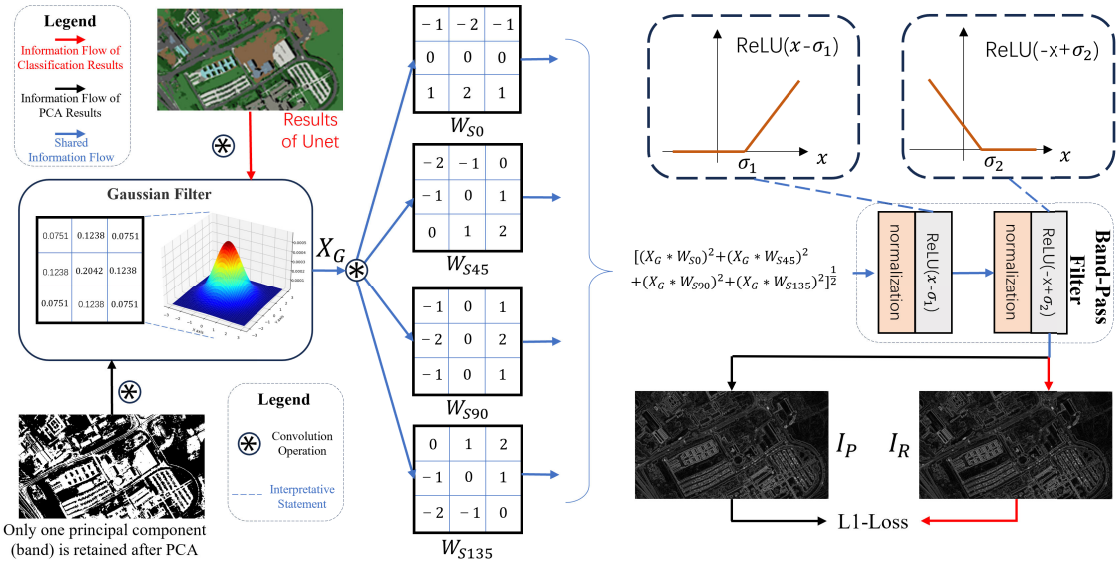


Fig. 6. Algorithm procedure of ECM based on the multidirectional Sobel operator. I_R and I_P represent the edge information images of the two information flows, respectively.

accurate pairing with global features. Finally, the classification results of the branches are output by SCGFM.

E. Spectral Feature: Pixel-Net

The pixel-level information of the HSI can be viewed as a sequence of spectra in vector form. As an alternative algorithm to MLP, FKAN is also suitable for feature mining and analysis of sequence information. Alternatively, 1-D convolutions can be used to mine sequence information through weight sharing. Based on the above FKAN and its convolution principle, we design a simple yet effective Pixel-Net. The branching structure of this network is shown in Fig. 4, which contains 1-D convolutional layers based on FKAN. Considering the better ability of FKAN to fit nonlinear processes, some dropout operations and a residual connection structure are introduced in this branch. This can enhance the generalization ability of the Pixel-Net to some extent. The output of this branch consists of two components. The first component serves as input to the subsequent CSGFM, which interacts with the patch-level features to adaptively learn multiple feature weights for fusion. The second component is fed into the classifier, thereby constraining the learned branch directions and facilitating the control of spectral information weights within the model.

It is worth noting that the nonlinear convolution method of FKAN can effectively improve the feature extraction ability of individual convolutional processes compared to regular linear convolutional kernels. Therefore, Patch-Net and Pixel-Net can achieve competitive performance with a small number of network layers and channels. These two branches of the network are designed with a small number of nodes, channels, and network layers. This is also one of the important reasons for the lightness and efficiency of the proposed model.

F. CSGFM

More complete information representation provides a more adequate feature exploration method for the model. However, different types of features, and in particular different scales of features, face differences in semantic information when fused. In HSI classification tasks, pixel-level vector features focus more on spectral information, small-scale samples contain more local spatial and spectral details, and features of large-scale samples have more global vision. Therefore, in order to avoid information loss, it is necessary to ensure that details and global information do not interfere with each other.

Features at different scales may contain duplicated or even conflicting information, which, if fused directly, can lead to information redundancy or mutual interference. Our proposed CSGFM achieves accurate pairing of small-scale to large-scale features and alleviates potential redundancy in the fusion process. In addition, the CSGFM avoids the feature interference problem caused by direct and indirect fusion and achieves full cooperative utilization of pixel-, patch-, and global-level features.

The mechanism consists of three parts: input construction, module structure, and final fusion under constraints at each end. The input to this mechanism requires three branches to be constructed jointly. Similar to the query in the self-attention mechanism, the output features of the Pixel-Net are used as the "guide" (G) to retrieve features of the Patch-Net and ECU-Net and guide them to fuse with the corresponding regional features of each stage of ECU-Net according to weight. At the ECU-Net, pixel- and patch-level feature vectors corresponding to spatial locations are provided for accurate pairing to mitigate the effects caused by inconsistent semantic information. The CSGFM is designed based on the self-attention module. While retaining the exploration method for key (K) in the self-attention mechanism, other homologous network branches are used for queries, and the other two homologous network branches are guided to integrate, enabling multiscale feature interactions. The specific implementation process is shown in Fig. 7. First, the spatial position of the sample used in the Patch-Net is located, and the feature data of the corresponding

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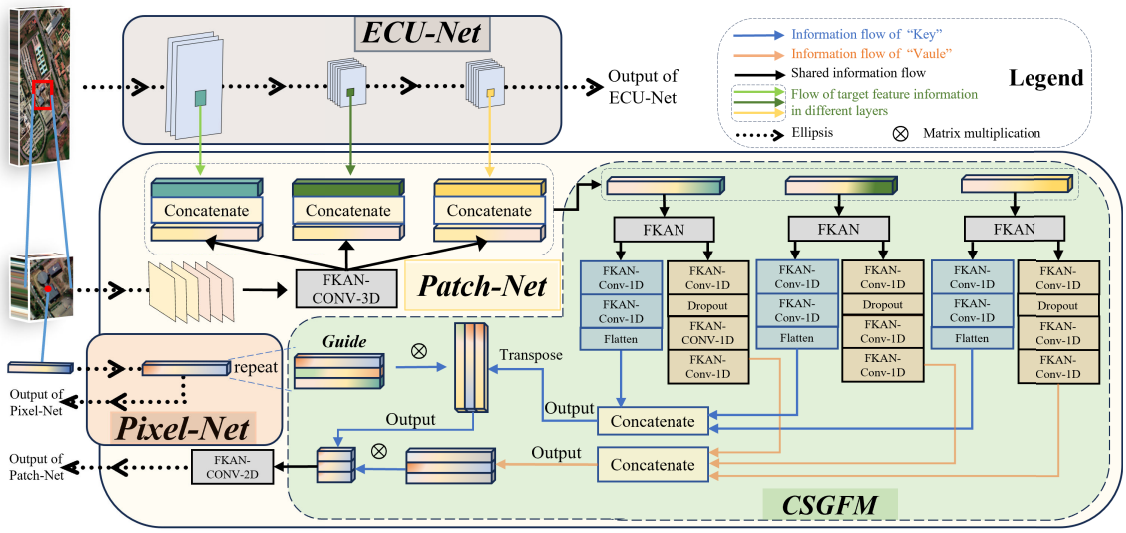


Fig. 7. Detailed structure and information flow of CSGFM. The inputs to this module comprise intermediate features derived from the three branches. Pixel-Net supplies features that guide fusion, while Patch-Net and ECU-Net produce features for fusion through precise pairing.

location of each encoder are taken in the ECU-Net. Second, the feature data were concatenated with the feature vectors in the Patch-N and fed into the multilayer KAN to generate K and the value (V) of each level. Then, K and V are stacked at each level to generate the feature tensor. Finally, G performed matrix multiplication with K at all levels to obtain the weight and used it to guide the addition of V. The proposed module realizes the fusion of these three scale features. In addition, in order to constrain the learning direction of G, K, and V, we supervise the output features of the branches at both ends of the scale range. That is, the outputs of Pixel-Net and ECU-Net are fed into their respective classifiers to construct the final classification result. This approach enables quadratic corrections to the learning direction for each branch.

IV. Result

This section presents relevant experimental validation of the proposed model, including comparison experiments with other models, critical parameter experiments, and ablation experiments. In addition, the datasets used and the detailed experimental setup are presented. Finally, the experimental results of each section are analyzed and discussed.

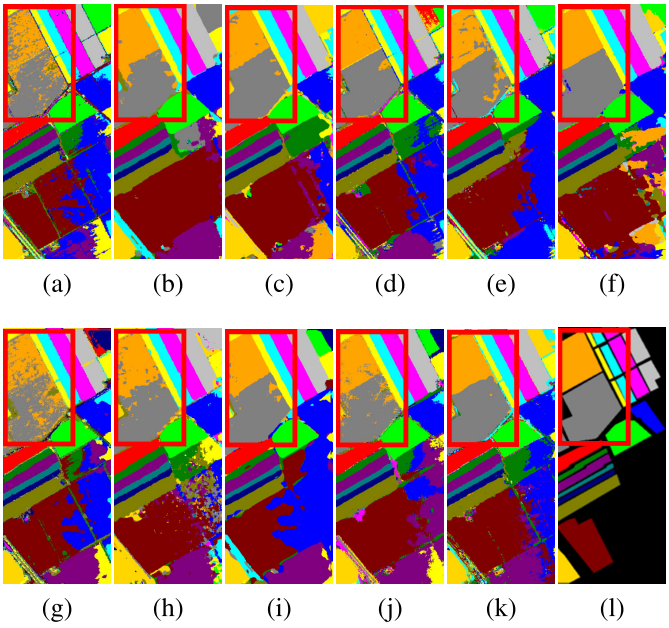


Fig. 8. Classification maps of different methods for the SV dataset. (a) SVM. (b) MVAHN. (c) MRViT. (d) DSGSF. (e) FM. (f) Mamba. (g) HKAN. (h) SSMLP. (i) CLOL. (j) ACB. (K) HPFN. (l) Ground truth.

A. Datasets

In this article, the performance of the proposed model is evaluated on four widely used HSI datasets, including the Pavia University (PU) dataset, the Indian Pines (IP) dataset, the Salinas Valley (SV) dataset, and the GRSS-DFC-2013-Houston (HS) dataset. The PU dataset and the HS dataset belong to urban scenes, and the IP and SV datasets are farmland scenes.

1) *PU:* This dataset is the distribution information of hyperspectral ground-based objects captured over the PU and contains nine labeled ground-based objects in total. The spatial range consists of 640 × 340 pixels and 102 bands in the spectral range of 430–680 nm (after excluding the interference band). The spectral and spatial resolutions are 1.3 m per pixel (mpp) and 3.74 nm, respectively.
2) *SV:* This dataset contains remote sensing HSI of ground objects and corresponding labeled information for 16 categories in the SV, California. The SV dataset has a spatial size of 512 × 217 and a spatial resolution of 3.7 mpp. After correction and processing, 204 spectral bands remain in this dataset.
3) *IP:* This dataset also belongs to the farmland scenes, which collects remote sensing hyperspectral ground-based object information in the northwestern Indiana region. The IP dataset has a total of

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TABLE I
DEFAULT SETTING OF PARAMETERS

Parameters	Default setting	Parameters	Default setting
batch_size	256	n_{L1}	3
weight_decay	0.0001	σ_1	0.4
patch_length	7*7	σ_2	0.45
PCA(Patch-Net)	16 channels	ζ	1
η	1	γ	1
k_{L1}	0.2	β	0.8

145 × 145 pixels and contains 16 different labeled classes. The dataset covers a spectral range from 400 to 2500 nm, with a spectral resolution of 10 nm and a spatial resolution of 20 mpp. This dataset eventually retained 200 spectral bands after removing the water absorption bands.

4) *HS:* This dataset was distributed for the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Image Analysis and DataFusion (IADF) Contest [48]. The HS dataset is a more challenging urban dataset, which contains 15 different labeled ground-based objects at the spatial scale of 349 × 1905 pixels. The spectral coverage ranges from 0.38 to 1.05 μm with a spatial resolution of 2.5 m.

B. Experimental Setup

To validate the performance of the proposed model, we evaluate the classification accuracy, the number of parameters, and the amount of computation and compare it with some representative HSI classification models. During the training process of 150 iterations, the initial value of the learning rate is set to 0.005, and the learning rate is adjusted in a stepwise manner, that is, it decays to its original value of 0.1 every 30 iterations of training. Hundred samples of each labeled class were randomly selected for training (50 samples were used for the HS dataset), and one-fifth of the training set was taken for validation. When the total number of samples does not satisfy the above requirements, half of the total samples of this class are taken. Finally, the remaining samples form a test set to evaluate the performance of the model. The detailed parameters (take the PU dataset for example) of the model construction are given in Table I. The choice of ζ, η, and γ is influenced by the dataset. It supports manual adjustment of the three-branch weights in different scenarios to flexibly adapt to the requirements of EPI and classification accuracy. The other hyperparameters do not vary with the dataset.

The comparison models selected in the experiments are SVM, multiple vision architectures-based hybrid network (MVAHN) [49], mixed residual convolutions with vision transformer (MRViT) [50], dual-view spectral and global spatial feature fusion network (DSGSF) [31], full model (FM) [30], HKAN [45], spatial-spectral multilayer perceptron (SSMLP) [51], Mamba [52], CLOL [53], and ACB [54]. These models cover pixel-, patch-, and global-level networks. In addition, these models include the different scales' integration model (DSGSF), the complete information representation model (FM), the novel MLP network (SSMLP), the recently emerging ViT models (MRViT and MVAHN), Mamba, and KAN (HKAN). Except for the number of samples, the default settings in the relevant literature are used for the other model hyperparameters. In the comparison experiments, all models are evaluated by overall accuracy (OA), average accuracy (AA), kappa, and the edge preservation index (EPI). The EPI is calculated by extracting the gradients in the horizontal and vertical directions using the Sobel operator. The mathematical procedure is given as follows:

$$\mathrm{EPI} = \frac{2 \cdot \sum(\mathbf{G}_I \circ \mathbf{G}_O)}{\sum \mathbf{G}_I^2 + \sum \mathbf{G}_O^2} \tag{8}$$

where $\mathbf{G}_I$ and $\mathbf{G}_O$ denote the classification map and the HSI (after PCA) gradient magnitude matrix, respectively. The gradient magnitude is the Euclidean norm of the gradients in both directions computed by the Sobel operator. $\circ$ denotes the Hadamard product. $\sum$ is used to calculate the sum of all elements in this matrix. Full-pixel classification maps of these models are output to compare visual differences such as oversmoothing and salt-and-pepper noise. To eliminate chance errors, all experiments were performed independently five times, and the results were averaged.

C. Result and Analysis

The experimental results are shown in Figs. 8–11 and Tables II–V. Among the four datasets, the SVM considering only spectral information has significantly lower OA, AA, and kappa than most models. It shows that the introduction of spatial information can effectively improve the classification accuracy of HSIs. However, there are certain classes in which SVM shows competitive performance (e.g., the bricks in the PU dataset and the water in the HS dataset). In fact, some models that take into account both spectral and spatial information have even lower classification accuracy than SVM in these classes. This suggests that some categories may be more sensitive to spectral information. In addition, the classification results of SVM have the clearest bounds among all models but also show the most significant salt-and-pepper noise. The visual performance of such classification maps is difficult to measure by a single evaluation criterion due to the different requirements of different tasks. In mapping and planning tasks, the sharpness of the boundary is often more important than the presence of noisy points in the region. In contrast, in vegetation area classification tasks, it is often necessary to classify interfering pixels within a certain category, such as land or weeds, into the same category. Since it is difficult for existing single networks to adapt flexibly to a variety of different classification tasks, we design an HPFN that retains more orthogonality of spectral and spatial manual assignments, which enhances the flexibility and compatibility of a single model under a variety of tasks.

As a hybrid network combining GCN, CNN, and ViT, MVAHN extracts HSI features from multiple perspectives. Both ViT and GCN focus more on large spatial scale information or relational information between local samples, thus potentially increasing the dependence of the model on spatial information. In comparison experiments, both hybrid model MVAHN and transformer-based MRViT achieve excellent classification accuracy. In datasets with farmland scenes, such as the SV dataset and the IP dataset, the fuzzing phenomenon in the classification maps of these two models seems to reduce

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TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE SV DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	MVAHN	VIT	DSGSF	FM	Mamba	HKAN	SSMLP	CLOL	ACB	HPFN
Brocoli_green_weeds 1	99.63	**100.00**	99.97	99.95	99.27	**100.00**	99.48	99.63	99.63	**100.00**	**100.00**
Brocoli_green_weeds 2	99.53	99.99	99.96	99.83	**100.00**	**100.00**	96.58	**100.00**	99.99	99.94	99.94
Fallow	99.63	99.75	**100.00**	97.86	99.89	**100.00**	98.72	99.95	99.95	**100.00**	**100.00**
Fallow_rough_plow	99.77	98.91	99.88	97.12	99.92	**99.92**	99.77	99.81	99.54	99.57	99.76
Fallow_smooth	96.51	**99.87**	99.41	99.55	99.65	98.75	99.57	99.40	99.63	99.76	99.84
Stubble	99.53	**100.00**	**100.00**	99.87	**100.00**	**100.00**	99.97	99.97	**100.00**	99.96	99.95
Celery	99.37	99.99	**100.00**	99.68	99.37	99.94	98.53	99.57	99.60	99.91	99.97
Grapes_untrained	81.18	96.59	96.22	99.62	94.37	**99.53**	78.52	93.64	94.08	91.88	97.29
Soil_vinyard_develop	97.51	99.75	99.98	99.60	99.77	**100.00**	97.97	99.83	99.89	99.92	99.93
Corn_senesced_weeds	93.61	98.88	99.18	94.68	99.69	99.65	94.12	**99.78**	98.79	98.63	99.68
Lettuce_romaine_4wk	95.35	99.93	98.57	96.06	99.48	**100.00**	96.90	99.74	99.74	**100.00**	99.80
Lettuce_romaine_5wk	99.89	99.97	99.89	99.66	99.84	**100.00**	97.92	99.92	99.92	99.97	99.52
Lettuce_romaine_6wk	98.28	99.94	**100.00**	99.23	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.53
Lettuce_romaine_7wk	94.54	99.97	99.09	96.75	99.79	**100.00**	99.18	99.90	99.90	**100.00**	99.60
Vinyard_untrained	69.00	90.04	**99.53**	95.04	86.02	99.25	64.09	87.58	90.58	94.08	98.95
Vinyard_vertical_trellis	99.41	**100.00**	**100.00**	97.66	99.18	99.88	96.31	99.30	99.41	50.32	98.16
OA(%)	90.54	97.72	98.96	98.39	96.71	**99.70**	89.25	96.81	97.27	97.26	99.18
AA(%)	95.17	98.97	99.34	98.26	98.52	**99.81**	94.85	98.63	98.79	98.96	99.59
Kappa	0.8945	0.9746	0.9884	0.9820	0.9633	**0.9967**	0.8800	0.9643	0.9695	0.9694	0.9909
EPI	0.5023	0.3977	0.3760	0.5364	0.5647	0.4750	**0.5312**	0.3797	0.5042	0.4475	0.5254

TABLE III

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE PU DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	MVAHN	MRVIT	DSGSF	FM	Mamba	HKAN	SSMLP	CLOL	ACB	HPFN
Asphalt	82.59	98.68	98.81	99.09	97.21	98.48	91.26	96.91	97.18	98.17	**99.30**
Meadows	89.82	**99.82**	99.78	99.59	96.57	98.66	85.92	97.00	98.57	94.74	99.74
Painted metal sheets	84.04	96.61	96.05	94.42	96.85	99.30	78.94	92.55	**99.90**	98.67	99.58
Gravel	96.32	96.85	97.65	98.83	99.22	97.36	96.76	96.32	97.91	99.10	**99.31**
Trees	99.36	99.51	99.87	97.72	99.76	**100.00**	99.92	**100.00**	**100.00**	**100.00**	99.97
Shadow	88.25	99.34	98.96	96.78	97.04	**100.00**	83.81	99.09	99.80	98.98	99.88
Bare Soil	95.04	95.23	96.53	96.89	99.59	**100.00**	89.11	99.43	**100.00**	99.90	99.93
Bitumen	86.54	92.85	93.63	94.99	92.55	98.57	78.56	95.51	**98.77**	98.11	97.35
Bricks	99.88	96.24	98.01	99.02	**100.00**	**100.00**	96.93	99.53	98.94	99.54	99.96
OA (%)	89.05	98.40	98.53	98.32	96.83	98.82	87.04	97.06	98.62	96.89	**99.46**
AA (%)	91.32	97.24	97.70	97.48	97.64	99.15	89.02	97.37	99.01	98.58	**99.45**
Kappa	0.8563	0.9787	0.9804	0.9776	0.9579	0.9843	0.8306	0.9609	0.9817	0.9582	**99.28**
EPI	0.5538	0.4730	0.4629	0.5044	0.5477	0.5113	**0.5818**	0.4445	0.4504	0.4373	53.26

TABLE IV

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE IP DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	MVAHN	VIT	DSGSF	FM	Mamba	HKAN	SSMLP	CLOL	ACB	HPFN
Alfalfa	82.43	89.37	99.13	**100.00**	99.82	96.63	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Corn-no till	76.69	**98.69**	96.47	98.65	96.61	95.93	51.66	98.50	88.86	97.71	97.76
Corn-min till	76.51	94.20	95.92	98.41	97.51	99.73	65.75	98.93	98.49	**100.00**	97.93
Corn	88.74	93.14	92.85	81.25	99.98	**100.00**	94.16	90.63	**100.00**	98.32	98.75
Grass/Pasture	93.38	98.12	99.30	98.59	**99.91**	95.25	82.77	99.16	96.61	99.59	99.33
Grass/Trees	91.19	93.18	99.31	96.30	99.88	**100.00**	91.75	98.15	**100.00**	97.69	99.42
Grass/pasture-mowed	86.11	98.67	96.47	86.67	99.84	**100.00**	**100.00**	93.33	**100.00**	**100.00**	95.02
Hay-windrowed	97.06	99.55	99.55	**100.00**	98.59	**100.00**	97.88	**100.00**	**100.00**	99.58	99.87
Oats	**100.00**	**100.00**	87.47	**100.00**	99.94	**100.00**	60.00	**100.00**	**100.00**	**100.00**	**100.00**
Soybeans-no till	81.24	90.83	89.40	90.45	**98.80**	97.89	66.51	94.48	90.25	98.12	97.94
Soybeans-min till	68.87	98.83	99.11	**99.28**	95.48	98.56	59.75	95.54	92.02	95.74	97.97
Soybeans-clean till	79.80	98.01	97.15	95.85	98.44	98.98	74.04	96.91	93.91	97.31	**98.99**
Wheat	97.68	99.19	99.73	99.01	**100.00**	**100.00**	98.10	99.50	**100.00**	**100.00**	99.95
Woods	92.93	99.68	99.69	**99.91**	99.56	99.56	91.24	99.83	99.14	99.70	99.59
Bldg-Grass-Tree-Drives	71.48	94.37	91.38	95.29	96.51	**100.00**	87.06	97.30	99.14	97.93	99.11
Stone-steel towers	93.00	92.96	96.54	84.91	98.38	96.06	91.49	92.45	**100.00**	95.74	97.36
OA (%)	80.56	96.87	97.01	97.32	97.08	98.43	71.60	97.27	94.56	97.62	**98.44**
AA (%)	86.07	96.17	96.22	95.29	**98.70**	98.66	82.01	97.17	97.41	98.59	98.68
Kappa	0.7803	0.9641	0.9657	0.9689	0.9666	0.9818	0.6776	0.9685	0.9372	0.9712	**0.9819**
EPI	**0.6512**	0.5396	0.5478	0.6022	0.5591	0.5287	0.5821	0.5899	0.5301	0.5682	0.5613

the noise, thus improving the classification accuracy. However, in urban datasets, excessive smoothing results in blurred edges of targets such as streets, residential areas, and railways, which adversely affects tasks such as mapping and planning.

Mamba is a novel model that has been widely used in image processing research in recent years. As shown in Tables III–V, Mamba shows outstanding classification performance on the IP dataset and achieves the best classification accuracy with a clear advantage on the SV dataset. This indicates that Mamba can effectively handle the HSI high-precision classification task. However, Mamba still has a lot of room for improvement in urban scenarios such as the

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TABLE V
QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE HS DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	MVAHN	VIT	DSGSF	FM	Mamba	HKAN	SSMLP	CLOL	ACB	HPFN
Healthy grass	93.84	**99.37**	98.48	97.33	89.18	94.71	85.58	96.50	92.09	92.77	98.29
Stressed grass	98.75	**99.91**	99.41	98.96	88.87	97.24	94.02	99.83	99.74	94.21	99.29
Synthetic grass	99.38	**100.00**	**100.00**	99.56	**100.00**	99.84	99.50	**100.00**	99.33	99.60	99.89
Trees	92.46	96.92	**99.83**	98.54	95.90	96.03	99.04	99.50	97.38	94.25	98.51
Soil	97.65	95.89	**100.00**	97.85	93.71	99.83	99.21	**100.00**	99.91	98.94	99.71
Water	97.82	99.24	98.71	98.12	96.00	99.25	95.56	99.64	99.56	96.00	**99.62**
Residential	90.07	**99.02**	93.43	95.74	93.35	96.36	84.50	97.21	89.90	91.76	97.17
Commercial	54.19	**100.00**	94.67	97.59	78.64	87.08	86.54	91.04	93.09	91.95	95.88
Road	76.62	96.59	96.05	96.73	86.19	90.44	84.64	90.02	95.05	90.68	**97.41**
Highway	78.93	99.12	**99.66**	98.08	93.63	99.91	84.56	99.24	96.98	98.73	99.50
Railway	63.46	99.01	98.38	98.19	94.35	92.60	91.01	**99.83**	97.00	95.17	98.10
Parking Lot 1	54.35	93.89	98.12	97.94	86.81	95.91	91.17	96.37	95.68	96.42	**98.64**
Parking Lot 2	49.40	91.18	97.76	95.76	99.51	94.38	96.21	**99.52**	95.39	94.80	98.44
Tennis Court	97.09	97.57	99.75	98.23	**100.00**	**100.00**	98.48	**100.00**	**100.00**	**100.00**	99.82
Running Track	98.52	96.35	100.00	99.22	99.84	98.83	98.93	**100.00**	**100.00**	**100.00**	99.74
OA (%)	81.66	97.73	98.07	97.79	91.58	95.58	91.27	97.42	96.17	95.02	**98.45**
AA (%)	82.84	97.60	98.28	97.86	93.07	96.16	92.60	97.91	96.74	95.69	**98.67**
Kappa	0.8016	0.9755	0.9791	0.9761	0.9089	0.9521	0.9055	0.9721	0.9585	0.9459	**0.9832**
EPI	0.3627	0.2391	0.2724	0.3565	0.2885	0.3556	**0.4970**	0.2753	0.2467	0.3388	0.3801

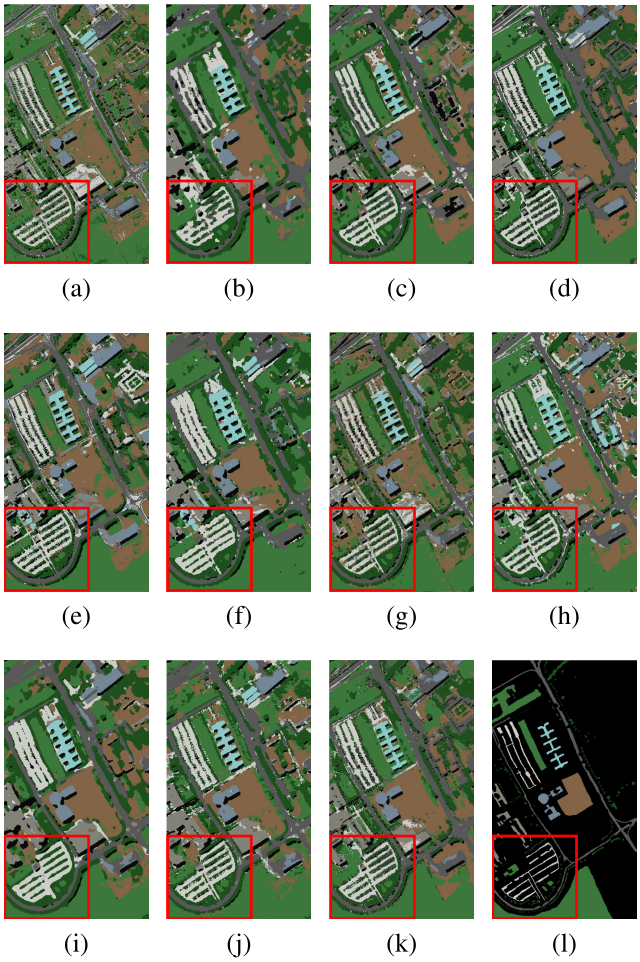


Fig. 9. Classification maps of different methods for the PU dataset. (a) SVM. (b) MVAHN. (c) MRViT. (d) DSGSF. (e) FM. (f) Mamba. (g) HKAN. (h) SSMLP. (i) CLOL. (j) ACB. (K) HPFN. (l) Ground truth.

HS dataset and the PU dataset. This not only reflects Mamba's ability to mine relational information between samples but also shows its dependence on spatial information similar to the transformer and GCN models. The proposed HPFN achieves suboptimal classification results on the SV dataset while maintaining a certain degree of edge sharpness. This

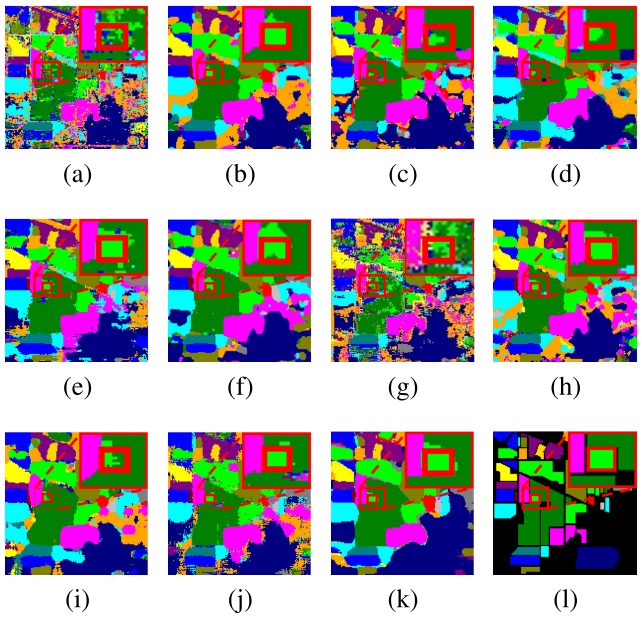


Fig. 10. Classification maps of different methods for the IP dataset. (a) SVM. (b) MVAHN. (c) MRViT. (d) DSGSF. (e) FM. (f) Mamba. (g) HKAN. (h) SSMLP. (i) CLOL. (j) ACB. (K) HPFN. (l) Ground truth.

demonstrates the comprehensive control of spatial and spectral information by the HPFN. The classification results of HPFN still have room for improvement as the spatial information weights are further increased in subsequent critical parameter experiments. Correspondingly, the classification effect of edges also decreases. Therefore, in this comparison experiment, on the one hand, HPFN has a slightly lower ability to extract spatial information than Mamba. On the other hand, given the comprehensive performance of the models, the enhancement of spectral information also sacrifices some of the classification accuracy.

DSGSF fuses U-net and LSTM to take into account both spatial and spectral information. Experimental results on four datasets show that pixel- and global-level fusion methods can maintain the sharpness of object boundaries and achieve better classification accuracy. DSGSF combines the classification

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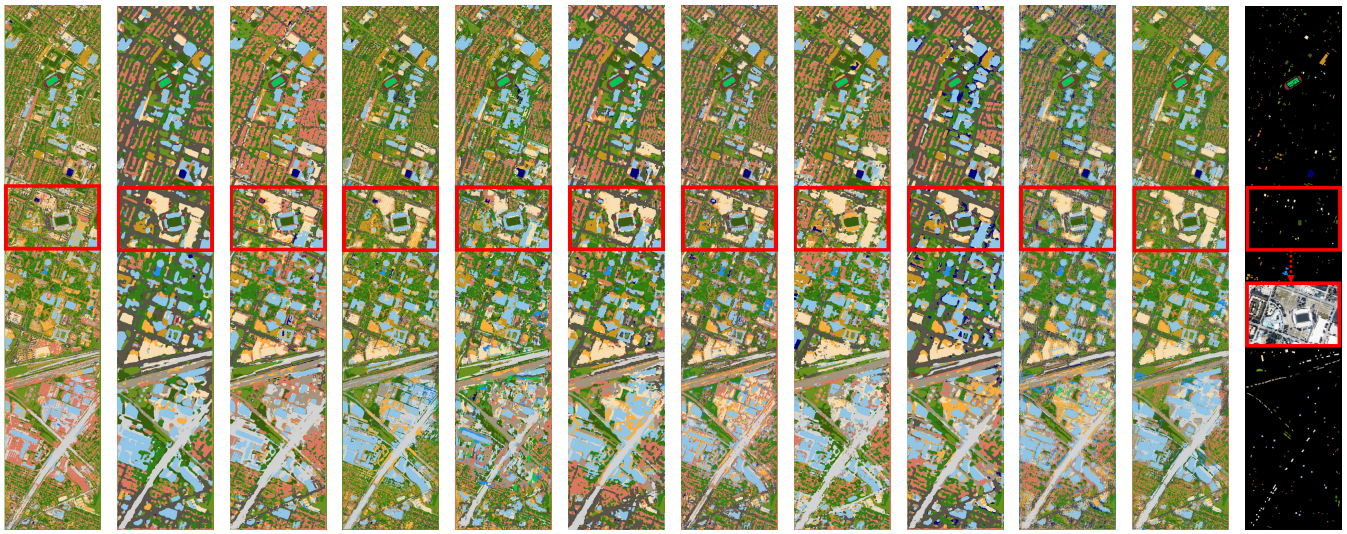


Fig. 11. Classification maps of different methods for the HS dataset. (a) SVM. (b) MVAHN. (c) MRViT. (d) DSGSF. (e) FM. (f) Mamba. (g) HKAN. (h) SSMLP. (i) CLOL. (j) ACB. (K) HPFN. (l) Ground truth.

results of the two networks only in the final stage, which retains higher orthogonality to the spectral–spatial information and facilitates the model to balance the two kinds of information. This model provides valuable enlightenment for this article to design a "pixel-locality-global" network. However, DSGSF neglects the interplay of intermediate features and the exploration of local information. Feature extraction methods based on U-net with 2-D CNN make DSGSF computationally expensive.

The application of KAN to deep learning tasks seems to support the design of more lightweight models. HKAN is a KAN convolutional hybrid HSI classification network with multiple dimensions. Experimentally, the network achieves good classification accuracy using only a small number of convolutional layers, pointing toward a viable solution for complex models with more complete information processing. However, there is a gap between HKAN and state-of-the-art models on the four datasets used, which may be caused by its shallow number of network layers. Moreover, as a novel learning algorithm, KAN is mostly used to solve mathematical or physical problems, and extensive cross-application studies are still lacking. The network itself also contains various hyperparameters that need to be determined manually, such as the number of grids and orders. Therefore, it remains to explore a more appropriate hyperparameter selection range for the HSI classification task. In this article, we use FKAN-based convolutional methods to integrate multiple levels of information and maintain competitive classification accuracy while reducing computational and memory costs. Considering that KAN can be used as an alternative to MLP for feature extraction, the SSMLP model is included as a comparison. In the experiments on the four datasets, SSMLP still underperformed the best model by about 1%–2%. This may be due to the fact that the feature extraction framework of this model mainly consists of a series of simple fully connected layers, which lacks sufficient nonlinear feature sensing capabilities. Moreover, the model also shows oversmoothed classification results, indicating that SSMLP still has the potential to improve the balance of spatial and spectral information utilization. Our proposed model employs FKAN-based convolutional methods, which are more focused on the complex nonlinear feature learning of HSI. With the cooperation of multiple scale branches and the ECM, the spatial–spectral information weights of the model can be flexibly adjusted to adapt to multiple task requirements.

Both CLOL and ACB are CNN-based lightweight models. Experimental results show that the performance of the two models is close. Due to the use of fewer learnable parameters and the lighter network structure, it is difficult to take into account and balance different information. This is one of the important reasons that limit the performance of lightweight models. To avoid these problems, HPFN adopts a more comprehensive network structure and introduces a lightweight FKAN. It effectively reduces the computational cost while improving the performance of the model.

Considering the incomplete information representation in most current models, FM models efficiently extract features from pixel- and patch-level data, respectively, and use multi-head self-attention to learn local-to-global relationships. Of all the compared models, FM achieves superior classification accuracy on the four datasets and takes the marginal classification cases into account to some extent. This suggests that a more complete representation of the information can effectively improve the classification accuracy while retaining the boundary information. However, FM does not account for the risk of information redundancy and potential feature conflicts that can occur in cross-scale fusion. Moreover, models dealing with complete information need to specifically extract features at different scales, which imposes a large computational cost on the model.

In addition, we quantitatively evaluate the edge classification effect of different model classification maps by EPI.

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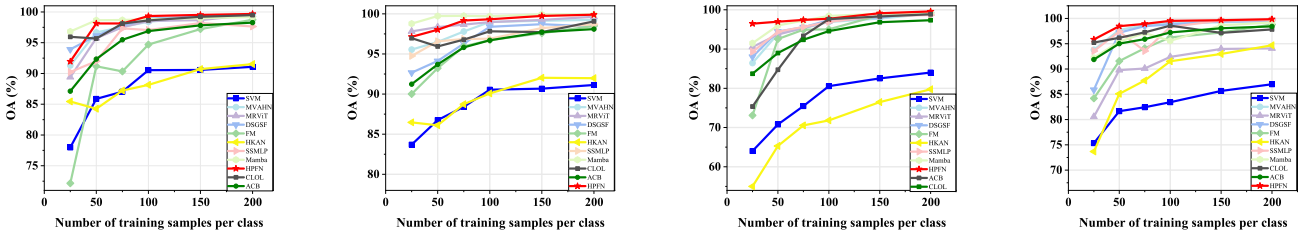


Fig. 12. Classiffcation accuracy of different methods under different scales of training samples. (a) PU. (b) SV. (c) IP. (d) HS.

HKAN achieves the best EPI on PU, SV, and HS datasets. This reflects the strong nonlinear feature extraction capability of HKAN with respect to spectral response. The SVMS using only spectral information also achieves a high EPI. However, Mamba, which relies on spatial information, can hardly achieve good boundary classification results. This is especially true on the SV dataset, although Mamba achieves the best classification accuracy while having the lowest EPI. The experimental results further verify that different datasets need to rely on different weights for spatial and spectral information. HPFN can achieve competitive classification accuracy while maintaining better EPI. In addition, the adjustable three-branch structure enables HPFN to adjust the boundary classification effect according to different task requirements.

Our proposed model achieves the best classification performance on the PU, IP, and HS datasets while achieving suboptimal results on the SV dataset. These experimental results demonstrate the ability of our model to perform HSI classification with high accuracy and competitive or better performance than current representative models. Combining the experimental results of FM and DSGSF, we validate the effectiveness of the proposed network model in improving classification accuracy and the visual effect of boundaries. In addition, our designed ECM structure and CSGFM make the model more flexible to cope with different scene requirements. On the urban datasets (HS and PU datasets), these two modules enable the proposed model to retain the gradient change information of the boundaries when it achieves the best classification accuracy. To compare the classification performance of different models more comprehensively, we set different numbers of samples for training on the four datasets. The experimental results are shown in Fig. 12, where SVM achieves the lowest classification performance in most cases. This again demonstrates the role of spatial information in improving the classification performance of HSIs. Experiments show that our model achieves competitive or better classification accuracy for different numbers of samples in multiple datasets. The results show that the feature extraction algorithm with KAN-convolution and CSGFM can efficiently perform the high-precision HSI classification task.

As shown in Table VI, although HKAN has a lower computational cost, its accuracy still has a lot of room for improvement. HPFN is significantly less computationally and parametrically demanding than most other deep learning models. In particular, the efficiency advantage of HPFN is fully reflected in datasets where the total number of pixels is small.

TABLE VI
COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS

	FLOPS(G)				Params (M)			
Dataset	PU	SV	IP	HS	PU	SV	IP	HS
MVAHN	5.25	7.93	7.83	27.56	0.25	0.38	0.37	0.36
VIT	684.86	717.72	717.72	717.72	27.06	28.07	28.07	28.07
DSGSF	8.46	4.60	0.89	27.16	0.39	0.40	0.40	0.40
SSMLP	73.69	73.69	73.69	73.69	0.85	0.86	0.86	0.86
FM	160.30	103.25	103.22	102.86	14.50	17.83	17.70	15.85
HKAN	0.09	0.09	0.09	0.09	0.022	0.022	0.022	0.022
MAMBA	20.04	12.17	2.29	69.65	0.18	0.19	0.19	0.18
ACB	0.10	0.10	0.10	0.10	0.018	0.022	0.022	0.021
CLOL	0.16	0.23	0.22	0.19	0.0035	0.0053	0.0052	0.0043
HPFN	4.03	2.43	0.78	12.45	0.051	0.053	0.053	0.052

For example, on the PU dataset, the computational complexity of FM is about 40 times that of HPFN. On the IP dataset, SSMLP is about 94 times more computationally demanding than HPFN, while on the HS dataset, this is reduced to nine and six times. This is due to the inclusion of the U-net architecture in our proposed model, whose computation in the encoding process increases as the input increases. The memory cost required by the HPFN is also greatly reduced compared to most other models, being about one-tenth or even less than that of the models. As a result, HPFN not only achieves competitive classification accuracy but also effectively reduces the computational cost requirement, which facilitates practical application and deployment in system-on-chip. However, due to its more complete network structure and competitive classification performance, the computational cost of HPFN is still higher than that of lightweight models (ACB and CLOL) that only use a single feature form. This is also an important direction for further optimization of the model in the future.

D. Ablation Experiments and Critical Parameters

To gain deeper insights into the proposed model, we investigate the effect of two critical parameters: the branching weights and the L1-loss weight. Notably, both parameters can be adjusted to optimize visual outcomes based on varying task requirements. Accordingly, we conduct experiments separately on the farmland dataset (SV) and the urban dataset (UP).

Fig. 13 illustrates the effect of different branch weights on the model performance. As the weight of the Patch-Net increases, both the classification accuracy and the background marginal accuracy of the model improve. It is shown that the Patch-Net effectively balances spatial and spectral information more effectively than the other two branches. As the weight of Pixel-Net increases, the classification of edge details

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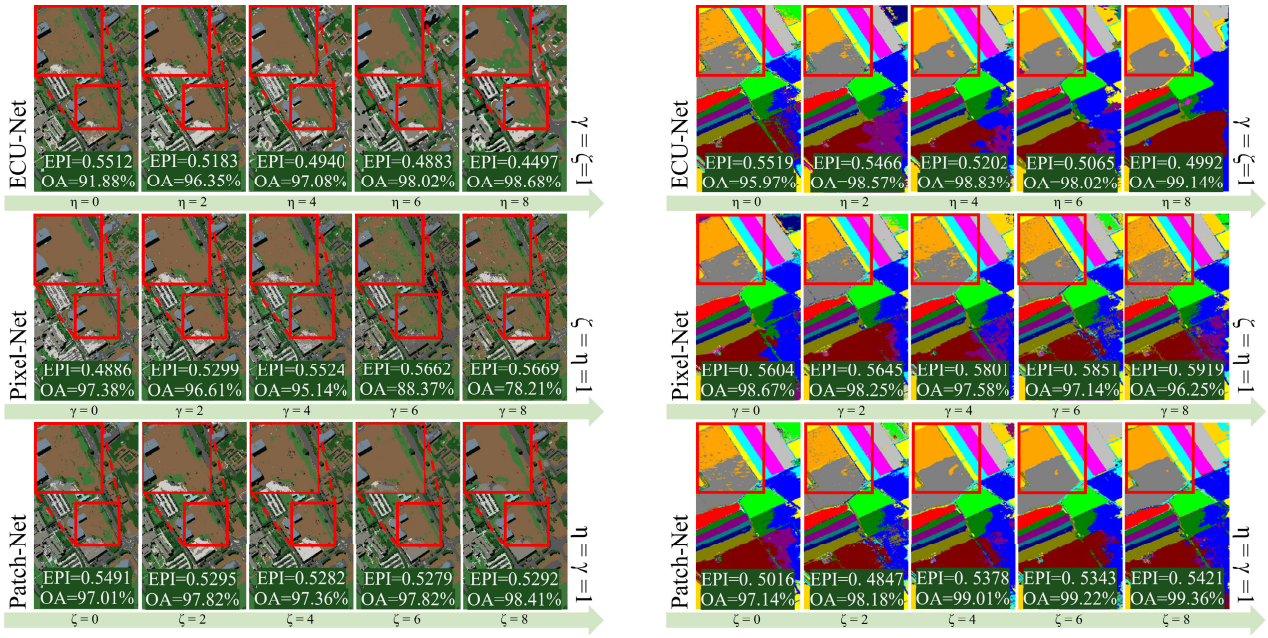


Fig. 13. Effects of weights η, ζ, and γ on classification results for different scenes. η, ζ, and γ denote the weights of the three branches, respectively. (a) PU dataset. (b) SV dataset.

becomes clearer; however, this is accompanied by an increase in salt-and-pepper noise and a decrease in OA. In turn, with higher weights assigned to the ECU-Net, the model exhibits improved classification accuracy, yet the oversmoothing problem becomes more pronounced. This phenomenon may be attributed to the enhanced spatial information, which leads to blurred results surrounding the target, especially since these objects tend to be clustered together, resulting in an optimistic classification accuracy. When only spectral information is utilized, the model struggles with the issue of spectral variability among surrounding objects, leading to the misclassification of neighboring objects into different classes and creating salt-and-pepper noise. In addition, the labeling of the dataset often involves categorizing objects by regions, which poses a risk of merging interspersed areas of different object categories into a single classification (e.g., grasslands in farmlands). Thus, some noisy points may represent correct classifications that have been inaccurately labeled.

For certain tasks, such as urban planning, clear boundaries are often more crucial than the presence of intraregion noisy points. In contrast, for the agricultural classification task, accuracy takes precedence. In practice, varying task requirements correspond to different model performance expectations. The proposed model takes into account diverse application scenarios by establishing three branches with more orthogonal information. HPFN allows manual tuning, maintaining competitive classification performance while ensuring lightweight design for ease of deployment. Similarly, to validate the effectiveness of the ECM in mitigating the potential oversmoothing problem, we tested its impact on edge sharpness and accuracy by adjusting the L1-loss weight. Fig. 14 shows that as the L1-loss weight increases, the object boundaries in the ECU-Net classification map gradually become clearer EPI

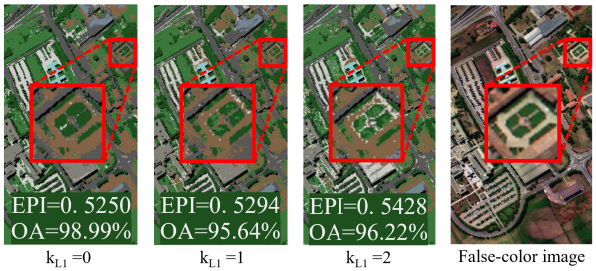


Fig. 14. Effect of L1-loss weight on classification results. K_{L1} denotes the weight of L1-loss.

gradually improves in ECU-Net classification maps although the classification accuracy decreases to varying degrees. Since the results of PCA originate primarily from spectral information, spectral variability may be a significant contributor to this phenomenon. Pixels within the same object category may exhibit substantial differences in spectral sequences. Through PCA and edge detection, these pixels can further disrupt the model's ability to accurately identify the corresponding object class.

The proposed model consists of three branches and two elementary modules. Given that the impact of the ECM has been evaluated in previous critical parameter experiments, we design ablation studies for the remaining four components to investigate the effect of these branches and modules. The experiment consists of separately removing the three branches and the cross-scale guided fusion module from the model. Table VII shows the results of the ablation experiment.

When only the Pixel-Net is removed, the model exhibits a decrease in classification accuracy for certain classes that are sensitive to spectral information. The removal of the Patch-Net

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TABLE VII
RESULTS OF ABLATION EXPERIMENT

	REMOVE	ECU-Net	Patch-Net	Pixel-Net	CSGFM	HPFN-MLP	None
UP	OA (%)	95.63	98.36	98.13	98.04	90.13	99.46
	AA (%)	96.93	98.63	97.73	98.54	91.30	99.45
	Kappa	0.9422	97.8143	0.9751	0.9739	0.8932	0.9928
SV	OA (%)	93.49	98.34	98.31	97.05	91.87	99.25
	AA (%)	97.09	99.05	99.18	98.68	95.97	99.61
	Kappa	0.9275	0.9815	0.9811	0.9672	0.9092	0.9916
IP	OA (%)	93.31	97.68	97.25	95.65	94.33	98.47
	AA (%)	95.50	97.48	98.07	98.24	97.38	98.66
	Kappa	0.9227	0.9730	0.9737	0.9497	0.9347	98.2282
HS	OA (%)	94.91673	96.51	96.71	96.77	94.96	98.48
	AA (%)	95.80	96.34	97.16	97.38	95.25	98.65
	Kappa	0.9417	0.9590	0.9644	0.9651	0.9455	0.9836

resulted in a further reduction of classification accuracy. Removing ECU-Net results in a significant deterioration of the classification performance of the model. As shown in Fig. 13, the visual classification effect varies when the fusion weights of the three branches are respectively reduced to zero. The removal of the ECU-Net yields the sharpest bounds, whereas the removal of the Pixel-Net leads to excessive smoothing. This experimental phenomenon partly reflects the antagonism and cooperation between spectral and spatial information. The model's reliance solely on spectral information hinders its ability to identify the occasional spectral variability of nearby objects of the same class, resulting in lower accuracy and sharper bounds. In turn, within a certain spatial range, the over-reliance on spatial information causes the model to incorrectly classify heterogeneous objects into the same category. This reliance resulted in an excellent performance on classification metrics but yielded a relatively blurred full-pixel classification map.

CSGFM is designed to facilitate the integration of features across different scales. In the experiments where this module was removed, the accuracy of the model decreased by about 1%. This decrease may be attributed to the direct fusion of features from different scales, leading to redundancy, conflict, and mutual interference among features. The CSGFM enhances the compatibility of features across different scales through accurate pairing of local and global features. Ablation experiments and critical parameter experiments validate the contributions and effects of each branch and module on the model performance, which enhances the comprehensive analysis of the proposed model.

Compared to the urban dataset, the classification performance is significantly affected when ECU-Net is removed from the farmland dataset. This suggests that the classification task on the farmland dataset is more dependent on spatial information. Correspondingly, on the SV dataset, the performance of the model drops by less than 1% after removing Pixel-Net. However, in the HS datasets with more complex environments, the classification accuracy decreases by about 2%. This verifies that there is a difference in the spatial–spectral dependence for different data sets. HPFN achieves a more flexible multi-information balance by exploring samples with different scales and adjustable weights.

In addition, HPFN-MLP is a replacement for all FKAN-based related content in HPFN with MLP-style operations. The classification accuracy of HPFN-MLP is significantly lower than that of HPFN. This indicates that FKAN has a stronger feature extraction capability than MLP for the same lightweight scale.

V. CONCLUSION

With the continuous development of deep learning and exploration of remote sensing data processing techniques, most current HSI classification models have achieved high performance. However, the transition to practical applications is a challenge that must be faced by HSI classification techniques and even the field of deep learning-based remote sensing image processing. Due to unavoidable issues such as insufficient label coverage and region partitioning of labels in existing HSI datasets, it is difficult to comprehensively evaluate the practical value and true performance of a model by classification accuracy alone. Motivated by this, we propose an HPFN based on FKANs, which aims to achieve high-accuracy classification while meeting the practical requirements of high efficiency, low cost, and tunability for boundary classification.

To take into account both accuracy and lightweight issues, we use the convolution method of FKAN to aggregate learnable parameters in the nonlinear fitting procedure. Moreover, we design more orthogonal and complete information exploration patterns with Pixel-Net, Patch-Net, and ECU-Net to achieve competitive classification accuracy with low operation costs. We develop a quadratic constraint for the global feature extraction process as ECM. By optimizing the loss function constructed between the boundary information of the original data and the classification map, the proposed module effectively mitigates the oversmoothing of the full-pixel classification map of ECU-Net. The design of the three branches of the model also allows manual tuning of the spectral and spatial weights. That is, the three branches depend on different information and the weights can be adjusted to meet the requirements of the scenario according to the actual application. To more fully validate the performance of the proposed model, we conduct extensive comparative experiments on four datasets with the addition of ablation experiments and evaluation of the computational cost. Experimentally, the effectiveness and performance of each module of the proposed model are verified.

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REFERENCES

[1] B. Xi et al., "DGSSC: A deep generative spectral–spatial classifier for imbalanced hyperspectral imagery," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1535–1548, Apr. 2023.

[2] J. Xu et al., "Fuzzy graph convolutional network for hyperspectral image classification," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107280. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197623014641

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[3] G. Yang, K. Huang, W. Sun, X. Meng, D. Mao, and Y. Ge, "Enhanced mangrove vegetation index based on hyperspectral images for mapping mangrove," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 236–254, Jul. 2022.

[4] J. Yuan, S. Wang, C. Wu, and Y. Xu, "Fine-grained classification of urban functional zones and landscape pattern analysis using hyperspectral satellite imagery: A case study of Wuhan," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3972–3991, 2022.

[5] B. Zhang, Y. Chen, Z. Li, S. Xiong, and X. Lu, "SANet: A self-attention network for agricultural hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501315.

[6] L. Windrim, A. Melkumyan, R. J. Murphy, A. Chlingaryan, and R. Leung, "Unsupervised ore/waste classification on open-cut mine faces using close-range hyperspectral data," *Geosci. Frontiers*, vol. 14, no. 4, Jul. 2023, Art. no. 101562. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1674987123000294

[7] D. Xiao, Q. H. Vu, B. T. Le, and T. T. L. Ha, "A method for mapping and monitoring of iron ore stopes based on hyperspectral remote sensing-ground data and a 3D deep neural network," *Neural Comput. Appl.*, vol. 35, no. 16, pp. 12221–12232, Jun. 2023, doi: 10.1007/s00521-023-08353-y.

[8] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[9] J. Plaza, A. Plaza, R. Perez, and P. Martinez, "On the use of small training sets for neural network-based characterization of mixed pixels in remotely sensed hyperspectral images," *Pattern Recognit.*, vol. 42, no. 11, pp. 3032–3045, Nov. 2009. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320309001496

[10] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[11] A. Agarwal, T. El-Ghazawi, H. El-Askary, and J. Le-Moigne, "Efficient hierarchical-PCA dimension reduction for hyperspectral imagery," in *Proc. IEEE Int. Symp. Signal Process. Inf. Technol.*, Dec. 2007, pp. 353–356.

[12] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.

[13] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[14] M. Sohail, Z. Chen, B. Yang, and G. Liu, "Multiscale spectral–spatial feature learning for hyperspectral image classification," *Displays*, vol. 74, Sep. 2022, Art. no. 102278. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0141938222000981

[15] J. Bai, W. Shi, Z. Xiao, T. A. A. Ali, F. Ye, and L. Jiao, "Achieving better category separability for hyperspectral image classification: A spatial–spectral approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9621–9635, Jul. 2024.

[16] K. Yang, H. Sun, C. Zou, and X. Lu, "Cross-attention spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518714.

[17] D. Wang, B. Du, and L. Zhang, "Spectral–spatial global graph reasoning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 12924–12937, Aug. 2024.

[18] M. Shi and J. Ren, "A lightweight dense relation network with attention for hyperspectral image few-shot classification," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 106993. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197623011776

[19] Z. Zhang, H. Zhou, C. Zhang, X. Zhang, and Y. Jiang, "A multi-level deformable gated aggregated network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 123, Sep. 2023, Art. no. 103482. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1569843223003060

[20] C. Zhao, W. Zhu, and S. Feng, "Superpixel guided deformable convolution network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3838–3851, 2022.

[21] M. Hu, C. Wu, and L. Zhang, "GlobalMind: Global multi-head interactive self-attention network for hyperspectral change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 465–483, May 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271624001539

[22] H. Cao, J. Cao, Y. Chu, Y. Wang, G. Liu, and P. Li, "Global-local manifold embedding broad graph convolutional network for hyperspectral image classification," *Neurocomputing*, vol. 602, Oct. 2024, Art. no. 128271. [Online]. Available: https://www.ciencedirect.com/science/article/pii/S0925231224010427

[23] K. Qu, C. Wang, Z. Li, and F. Luo, "Spatial–Spectral attention graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528317.

[24] G. Wu, M. A. A. Al-Qaness, D. Al-Alimi, A. Dahou, M. Abd Elaziz, and A. A. Ewees, "Hyperspectral image classification using graph convolutional network: A comprehensive review," *Expert Syst. Appl.*, vol. 257, Dec. 2024, Art. no. 125106. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417424019730

[25] S. Wan et al., "Multi-level graph learning network for hyperspectral image classification," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108705. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320322001868

[26] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[27] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.

[28] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "QTN: Quaternion transformer network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 12, pp. 7370–7384, Dec. 2023.

[29] H. Feng, Y. Wang, Z. Li, N. Zhang, Y. Zhang, and Y. Gao, "Information leakage in deep learning-based hyperspectral image classification: A survey," *Remote Sens.*, vol. 15, no. 15, p. 3793, Jul. 2023.

[30] J. Yang, B. Du, and L. Zhang, "Overcoming the barrier of incompleteness: A hyperspectral image classification full model," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14467–14481, Oct. 2024.

[31] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, May 2023, Art. no. 5512913.

[32] J. Wang and X. Tan, "Mutually beneficial transformer for multimodal data fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 12, pp. 7466–7479, Dec. 2023.

[33] J. Wang, M. Zhang, W. Li, and R. Tao, "A multistage information complementary fusion network based on flexible-mixup for HSI-X image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17189–17201, Dec. 2024.

[34] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.

[35] W. Han, W. Miao, J. Geng, and W. Jiang, "CMSE: Cross-modal semantic enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509814.

[36] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6804–6818, Nov. 2023.

[37] W.-S. Hu, W. Li, H.-C. Li, F.-H. Huang, and R. Tao, "Global clue-guided cross-memory quaternion transformer network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7357–7371, Apr. 2025.

[38] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231221004033

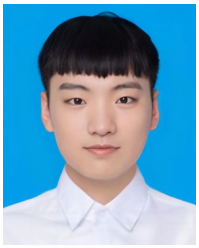
[39] Y. Lu, M. Gong, Z. Hu, W. Zhao, Z. Guan, and M. Zhang, "Energy-based CNN pruning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000214.

[40] R. Kemker, M. McClure, A. Abitino, T. L. Hayes, and C. Kanan, "Measuring catastrophic forgetting in neural networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Apr. 2018, pp. 1–12. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/11651

[41] K. Shukla, J. D. Toscano, Z. Wang, Z. Zou, and G. E. Karniadakis, "A comprehensive and FAIR comparison between MLP and KAN representations for differential equations and operator networks," *Comput. Methods Appl. Mech. Eng.*, vol. 431, Nov. 2024, Art. no. 117290. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0045782524005462

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[42] Z. Liu et al., "KAN: Kolmogorov–Arnold networks," 2024, *arXiv:2404.19756*.
[43] A. D. Blealtan. (2024). *Efficient-kan*. [Online]. Available: https://github.com/Blealtan/efficient-kan
[44] Z. Li, "Kolmogorov–Arnold networks are radial basis function networks," Tech. Rep., 2024. [Online]. Available: https://doi.org/10.48550/arXiv.2405.06721
[45] A. Jamali, S. Kumar Roy, D. Hong, B. Lu, and P. Ghamisi, "How to learn more? Exploring Kolmogorov–Arnold networks for hyperspectral image classification," 2024, *arXiv:2406.15719*.
[46] M. Kiamari, M. Kiamari, and B. Krishnamachari, "GKAN: Graph Kolmogorov–Arnold networks," 2024, *arXiv:2406.06470*.
[47] A. D. Bodner, A. S. Tepsich, J. N. Spolski, and S. Pourteau, "Convolutional Kolmogorov–Arnold networks," 2024, *arXiv:2406.13155*.
[48] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[49] F. Zhao, J. Zhang, Z. Meng, H. Liu, Z. Chang, and J. Fan, "Multiple vision architectures-based hybrid network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121032.
[50] Y. Cao, Y. Wang, Z. Yin, and Z. Yang, "Mixed residual convolutions with vision transformer in hyperspectral image classification," in *Proc. IEEE 22nd Int. Conf. Commun. Technol. (ICCT)*, Nov. 2022, pp. 1595–1599.
[51] Y. Sun, B. Liu, R. Wang, P. Zhang, and M. Dai, "Spectral–spatial MLP-like network with reciprocal points learning for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513218.
[52] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–Spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.
[53] C. Li, B. Rasti, X. Tang, P. Duan, J. Li, and Y. Peng, "Channel-layer-oriented lightweight spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504214.
[54] Y. Cui, L. Zhu, C. Zhao, L. Wang, and S. Gao, "Lightweight spectral–spatial feature extraction network based on domain generalization for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.



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